

A Practical Guide to Decision tree

TOPIC -- DECISION TREE

-- 01 --

Other techniques The above information is not where it ends for building and optimizing a decision tree. There are many techniques for improving the decision tree classification models we build. One of the techniques is making our decision tree model from a bootstrapped dataset. The bootstrapped dataset helps remove the bias that occurs when building a decision tree model with the same data the model is tested with. The ability to leverage the power of random forests can also help significantly improve the overall accuracy of the model being built. This method generates many decisions from many decision trees and tallies up the votes from each decision tree to make the final classification. There are many techniques, but the main objective is to test building your decision tree model in different ways to make sure it reaches the highest performance level possible.

-- 02 --

Are simple to understand and interpret. People are able to understand decision tree models after a brief explanation. Have value even with little hard data. Important insights can be generated based on experts describing a situation (its alternatives, probabilities, and costs) and their preferences for outcomes. Help determine worst, best, and expected values for different scenarios. Use a white box model. If a given result is provided by a model. Can be combined with other decision techniques. The action of more than one decision-maker can be considered. Disadvantages of decision trees:

-- 03 --

Now, we can use the formulas to calculate the phi function values and information gain values for each M in the dataset. Once all the values are calculated the tree can be produced. The first thing to be done is to select the root node. In information gain and the phi function we consider the optimal split to be the mutation that produces the highest value for information gain or the phi function. Now assume that M1 has the highest phi function value and M4 has the highest information gain value. The M1 mutation will be the root of our phi function tree and M4 will be the root of our information gain tree. You can observe the root nodes below

-- 04 --

Decision tree using flowchart symbols Commonly a decision tree is drawn using flowchart symbols as it is easier for many to read and understand. Note there is a conceptual error in the "Proceed" calculation of the tree shown below; the error relates to the calculation of "costs" awarded in a legal action.

-- 05 --

We will set D , which is the depth of the decision tree we are building, to three ($D = 3$). We also have the following data set of cancer and non-cancer samples and the mutation features that the samples either have or do not have. If a sample has a feature mutation then the sample is positive for that mutation, and it will be represented by one. If a sample does not have a feature mutation then the sample is negative for that mutation, and it will be represented by zero. To summarize, C stands for cancer and NC stands for non-cancer. The letter M stands for mutation, and if a sample has a particular mutation it will show up in the table as a one and otherwise zero.

-- 06 --

Runtime issues Decrease in accuracy in general Pure node splits while going deeper can cause issues. The ability to test the differences in classification results when changing D is imperative. We must be able to easily change and test the variables that could affect the accuracy and reliability of the decision tree-model.

-- 07 --

Drawn from left to right, a decision tree has only burst nodes (splitting paths) but no sink nodes (converging paths). So used manually they can grow very big and are then often hard to draw fully by hand. Traditionally, decision trees have been created manually – as the aside example shows – although increasingly, specialized software is employed.

-- 08 --

One major drawback of information gain is that the feature that is chosen as the next node in the tree tends to have more unique values. An advantage of information gain is that it tends to choose the most impactful features that are close to the root of the tree. It is a very good measure for deciding the relevance of some features. The phi function is also a good measure for deciding the relevance of some features based on "goodness". This is the information gain function formula. The formula states the information gain is a function of the entropy of a node of the decision tree minus the entropy of a candidate split at node t of a decision tree.

-- 09 --

Decision nodes – typically represented by squares Chance nodes – typically represented by circles End nodes – typically represented by triangles Decision trees are commonly used in operations research and operations management. If, in practice, decisions have to be taken online with no recall under incomplete knowledge, a decision tree should be paralleled by a probability model as a best choice model or online selection model algorithm. Another use of decision trees is as a descriptive means for calculating conditional probabilities. Decision trees, influence diagrams, utility functions, and other decision analysis tools and methods are taught to undergraduate students in schools of business, health economics, and public health, and are examples of operations research or management science methods. These tools are also used to predict decisions of householders in normal and emergency scenarios.

-- 10 --

Decision trees can also be seen as generative models of induction rules from empirical data. An optimal decision tree is then defined as a tree that accounts for most of the data, while minimizing the number of levels (or "questions"). Several algorithms to generate such optimal trees have been devised, such as ID3/4/5, CLS, ASSISTANT, and CART.

-- 11 --

This is the phi function formula. The phi function is maximized when the chosen feature splits the samples in a way that produces homogenous splits and have around the same number of samples in each split.

-- 12 --

The choice of node-splitting functions The node splitting function used can have an impact on improving the accuracy of the decision tree. For example, using the information-gain function may yield better results than using the phi function. The phi function is known as a measure of "goodness" of a candidate split at a node in the decision tree. The information gain function is known as a measure of the "reduction in entropy". In the following, we will build two decision trees. One decision tree will be built using the phi function to split the nodes and one decision tree will be built using the information gain function to split the nodes. The main advantages and disadvantages of information gain and phi function